Requirement Analysis

Project Title

AI Learning Assistant Using Emotion Detection and Google Gemini AI

Introduction

The AI Learning Assistant is an intelligent educational support system that detects the emotional state of learners from their text input and provides personalized learning guidance using Google Gemini AI. The system aims to improve student engagement and learning outcomes by understanding emotions such as confusion, frustration, curiosity, boredom, and confidence.

Problem Statement

Many online learning platforms provide the same learning experience to every student without understanding their emotional state. Students who feel confused or frustrated often lose motivation and require personalized guidance. This project addresses this issue by detecting learner emotions and generating adaptive AI-based responses.

Objectives

- Detect learner emotions from text input.
- Generate personalized learning suggestions.
- Improve student engagement.
- Reduce learner frustration.
- Provide an intelligent AI learning assistant.

Functional Requirements

- User enters learning-related text.
- Detect emotion using BERT and BiLSTM models.
- Generate personalized suggestions using Google Gemini AI.
- Display emotion prediction.
- Store session history.
- Display previous interactions.

Non-Functional Requirements

- Fast response time.
- User-friendly interface.
- Secure API integration.
- Reliable emotion prediction.

- Easy to maintain.

Software Requirements

- Python 3.10+
- Streamlit
- TensorFlow
- Transformers
- Google Gemini API
- Scikit-learn
- Pandas
- NumPy

Hardware Requirements

- Windows 10/11
- Minimum 8 GB RAM
- Intel i5 Processor or above
- Internet Connection

Technologies Used

- Python
- Streamlit
- Google Gemini AI
- TensorFlow
- Hugging Face Transformers
- BERT
- BiLSTM
- Pandas
- NumPy

Expected Outcome

The system successfully detects learner emotions and provides personalized AI-generated learning guidance, helping students improve their learning experience and motivation.